Network Intrusion Analysis Report

DataQualiQ | Cybersecurity decision support

Analysed filename: UNSW_NB15_testing-set.csv

Records analysed: 82332

Executive summary

82332 network records were analysed. 26700 were classified as Normal and 55632 as Malicious. The uploaded data received a quality score of 100/100 (Grade A). 5 records require Review; 0 were rejected.

Current-upload findings below are separate from saved official model evaluation.

Data quality

Quality score: 100/100; Grade A. Heuristic quality score, not prediction accuracy.

Data errors, quality issues and analytical/security anomalies are distinct. Counts can overlap; analytical anomalies are not automatically incorrect data or attacks.

Missing values (Quality issues): 0 affected records; 0 occurrences; 0 findings

Duplicate rows (Quality issues): 0 affected records; 0 occurrences; 0 findings

Invalid values (Data errors): 0 affected records; 0 occurrences; 0 findings

Inconsistencies (Quality issues): 0 affected records; 0 occurrences; 0 findings

Analytical/security anomalies (Analytical/security anomalies): 58543 affected records; 248284 occurrences; 32 findings

Field dur: 5868 occurrences

Field spkts: 10196 occurrences

Field dpkts: 8907 occurrences

Field sbytes: 9270 occurrences

Field dbytes: 12308 occurrences

Field rate: 6201 occurrences

Field sload: 6715 occurrences

Field dload: 18112 occurrences

Field sloss: 5499 occurrences

Field dloss: 11272 occurrences

Field details truncated for this report.

Duplicate counts exclude the first occurrence.

Warning: Unseen categories use saved encoding and require Review; no categories were learned.

Intrusion detection - current upload

Normal: 26700; Malicious: 55632; Rejected: 0

Normal percentage: 32.4%; Malicious percentage: 67.6%. Percentages use classified records only; rejected rows are excluded.

Average predicted-class probability: 90.8%

Model probabilities are estimates, not calibrated guarantees.

Model Performance - Official Test Evaluation

Saved official test evaluation: Phase 5 results from the reserved test dataset; NOT calculated from the current uploaded file.

UNSW-NB15 reserved official testing split; 82332 evaluated records

Accuracy: 86.4%; Precision: 80.7%; Recall: 99%; F1-score: 88.9%

Confusion matrix: rows actual, columns predicted; order Normal, Malicious.

True Normal: 26256; False Alarm: 10744

Missed Attack: 444; Correctly Detected Attack: 44888

False positives are normal traffic classified as malicious. False negatives are missed malicious traffic.

Reliability - predictive-input integrity

High: 82327; Review: 5; Low: 0; Rejected: 0 (included in Low).

High: prediction produced with no unresolved predictive-input integrity issue.

Review: prediction produced, but input had a relevant handled quality issue.

Low: prediction could not be safely produced due to invalid required input; rejected records have no class or probability.

Reliability describes input integrity, separate from model probability; it is not guaranteed correctness.

Suspicious / important records

Bounded selection: 20 important records, maximum 20, from 100 returned rows of 82332 total (offset 0). This is not a full-file ranking.

Order: Malicious + Review, Malicious + Low, rejected records, Malicious + High. Rejected status takes precedence over any supplied class.

Record numbers are original one-based CSV data-row positions, excluding the header. No extra pages were fetched.

Record 1: Malicious; predicted probability 86%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 2: Malicious; predicted probability 84.5%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 3: Malicious; predicted probability 78.2%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 4: Malicious; predicted probability 73.1%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 5: Malicious; predicted probability 74.8%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 6: Malicious; predicted probability 79%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

rate: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 7: Malicious; predicted probability 76.5%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 8: Malicious; predicted probability 74.2%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 13: Malicious; predicted probability 80.1%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 14: Malicious; predicted probability 76%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 15: Malicious; predicted probability 85.5%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 16: Malicious; predicted probability 76.9%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 17: Malicious; predicted probability 93.4%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

rate: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 18: Malicious; predicted probability 77.9%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 19: Malicious; predicted probability 84.2%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

rate: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 20: Malicious; predicted probability 84.1%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 21: Malicious; predicted probability 77.6%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 22: Malicious; predicted probability 95.4%; reliability High; No issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

Record 23: Malicious; predicted probability 71.3%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Record 24: Malicious; predicted probability 78.3%; reliability High; Issue detected

Reliability reason: Prediction produced with no unresolved predictive-field integrity issue. Analytical anomalies and preserved duplicates do not invalidate the input.

sload: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

smean: Requires five finite numbers, representable IQR fences, variation, and no identifier or binary-indicator interpretation.

Security recommendations

Backend decision-support recommendations for human review and investigation.

high priority: Prioritise human review of malicious predictions with input-integrity issues.

Reason: Malicious predictions requiring Review: 3. Input issues may affect interpretation.

Affected records: 3

high priority: Investigate model-predicted malicious records using independent security evidence.

Reason: 55632 records were predicted Malicious. A probability estimate does not establish a confirmed attack.

Affected records: 55632

medium priority: Preserve analytical anomalies for contextual security investigation.

Reason: 58543 distinct records have analytical findings; these are not automatically bad input or confirmed attacks.

Affected records: 58543

Limitations

Model probabilities are estimates, not calibrated guarantees. Input reliability is not guaranteed prediction correctness.

Analytical anomalies are not automatically attacks. The model requires compatible network features.

This prototype analyses uploaded network records, not live packet capture.

Only the currently returned record window is available for selection. The PDF does not include all predictions, evidence arrays, anomaly occurrences or raw audit structures.

Official evaluation is saved Phase 5 performance; model selection used that testing split, so it is not a fresh independent evaluation of the selected winner.

Analysis date/time is not supplied by the current backend response; no analysis timestamp was inferred.